

Explain the product. Qualify the evidence. Reduce the burden.

Codebase atlas, independent QA, granular specifications, semantic compression, dependency adoption and comparative product proof.

Planning cohort: 23 product/lab candidates, four pooled foundation homes and three supporting surfaces. Ten Tracera workstreams; product/assurance pairs elsewhere. PhenoApps and the absorbed Agentora capability are conditional additional plans.

Read this as a work program

This is not a fresh 52-repository audit or a product qualification. Source inspections and prior audit findings are inherited context. Product pilots are planned, not executed. No repository, deployment, engine state or lifecycle was mutated while generating the package.

Use the complete source tree

The ZIP contains the editable Markdown plans, subject dossiers, proposed requirements, provisional specification bundles, task graph, schema examples, reference checks and preserved contextual archives. START-HERE.md is the controlling reading entryptpoint.

Revision 1.1: ecosystem-first evolution

Reuse includes our own work. Improve shared capabilities with current and credible future consumers in view, and judge aggregate effects rather than repository-local convenience. Details: [architecture/ECOSYSTEM-FIRST-EVOLUTION.md](#).

Outstanding terminology

SROC/CDP have not been authoritatively defined in the retrieved context. Their names are retained without invented expansions. Unambiguous atlas and assurance work continues.

One connected understanding of each product

Explain, do not merely enumerate

A directory tree and API index are useful but insufficient. Connect exact source and artifact identity to user jobs, capabilities, interfaces, state, constraints, tests, evidence and dissatisfaction. Unknown purpose remains unknown; an agent must not invent historical rationale.

Account for every relevant artifact

Code, tests, native shells, scripts, configuration, data migrations, assets, prompts, generated files, vendor source and distribution inputs are all in scope. Use semantic units and source spans. Generated boilerplate inherits its producer contract; a critical three-line permission check may need several invariants.

Preserve authority

Tracera owns the persistent reconciled product model. Subject files retain their accepted source contracts. Actual work engines own transitions. Native runners produce test evidence. PhenoDocs renders views. A generated index or topic set cannot become a second mutable authority.

Reduce custom maintenance

Evaluate mature facilities and dependencies first, then narrow adapters or patches, then custom implementation where justified. Preserve ordering, errors, state, recovery, cancellation, compatibility, numerical properties and resource budgets. Fewer lines alone is not a win.

Keep product work moving

Use portable repo-local records before Tracera is complete. Close real useful slices while expanding the atlas. Do not wait for perfect documentation to fix a known defect, and do not call one useful repair the completion of the product.

Measure outcomes

Track verified/released/adopted capabilities, installed-CVP lead time, defects discovered after install, stale evidence, safe duplicate removal, maintenance burden and recovery interventions. Commits, tags and document volume are not substitutes.

Independent floors and credible failure detection

Coverage cells

Required unit, integration and E2E each retain at least 85% coverage in approved component/platform/profile/metric cells. Existing stricter policies remain. Lines, branches, functions and behavioral obligations are not interchangeable. Supplemental combined coverage cannot hide a weak required cell.

Denominators before numerators

Inventory eligible zero-hit units and freeze reviewed support/critical sets. Missing instruments, unsupported metrics and zero denominators are explicit limits, not 100%. Do not exclude difficult files, platforms or cases simply to pass.

Coverage is not a failure quota

All selected mandatory checks actually run and pass. Critical obligations are completely satisfied. Skips, empty selections, unavailable backends, malformed reports, timeouts and quarantined flakes do not count as passes. A known required defect cannot hide in the uncovered 15%.

Test the gate

Seed missing families, wrong revisions, changed denominator/critical sets, failed checks, mixed accumulators, broken links and malformed judge responses. Prove they propagate to non-acceptance. A schema and checksum establish only limited structure/integrity, not producer authenticity or domain truth.

Preserve qualitative evidence

Game/UI/audio quality needs explicit qualitative review alongside automation. Actual journeys must show the real artifact and backend. Branding can be rich and authored; it cannot impersonate implemented UI or performance evidence.

Reference tooling only

The package checker is intentionally limited. It always refuses to grant lifecycle approval. Its synthetic fixtures and measured test coverage concern the checker itself, not Civis, AgilePlus, Tracera or any other product.

Simplify without deleting the useful semantics

A useful refactor contract

Before changing a custom subsystem, record the outputs, errors, ordering, side effects, compatibility, permissions, persistence, timing and resources that must survive. Characterize actual useful behavior separately from bugs. Validate the actual consumers after replacement.

A dependency decision is broader than features

Compare functional and semantic fit, resource costs, testability, debugger behavior, supply chain, licensing, update/patch effort, operational burden and exit. Avoid wrappers that simply rename the full native API and become a second maintenance surface.

Patch discipline

Record upstream revision, owned delta, tests, owner, upstream issue where applicable, supported range and removal condition. A clean patch apply does not prove correct integration. Qualify upgrades and state compatibility.

Applet boundaries

An applet declares capabilities, ports, state, permissions, lifecycle, resources, surfaces and children. Composites can be applets. An applet is not automatically a repo, daemon or microservice. Use direct calls for suitable same-process paths and justify isolation/network overhead.

Dead paths

Accepted current behavior should be reconnected or implemented. Future experiments may retain specs without misleading executable stubs. Superseded/duplicate code needs consumer/provenance review. Optional/dynamic/external usage requires investigation. Unknown is not permission to delete.

Foundation consolidation

A future Pheno workspace can reduce repo overhead if packages keep precise ownership, interfaces, tests and release contracts. Do not replace a wide portfolio with one context-hostile universal product.

Our own work is upstream too

Reuse across both boundaries

Prefer use, modification, adaptation or a narrow wrapper over parallel handrolling, for external AND owned capabilities. Search our real packages, source owners, consumer contracts and prior extractions. Do not preserve a weak owned mechanism just because we wrote it.

Optimize the whole system

A repository assignment limits writes, not reasoning. Compare user value and total maintenance, dependency coupling, upgrade/patch work, build/runtime resources, migration and coordination. A local win that shifts greater burden or regressions downstream is not an ecosystem win.

Current and future consumers

Supported current consumers need preserved contracts and actual tests. Committed planned consumers need explicit requirements and appropriate contract spikes. Plausible future uses guide inexpensive seams and optionality, not speculative universal frameworks. Unknown consumers are not zero consumers.

One owner, bounded adapters

Improve the authoritative shared capability where appropriate. Keep product-specific policy at the proper boundary. Intentional separate implementations can be justified by different semantics or constraints. A future Pheno workspace is not one mandatory runtime, database or release train.

Every meaningful change carries impact

Record owned/external options, affected consumers, preserved behavior and resources, expected versus measured effects, tests, migration, rollback and review ownership in the existing work/PR/ADR. Keep purely local changes lightweight. A valid record is not an approval or parity proof.

Continue, do not perpetually rewrite

Reassess on new consumers, repeated workarounds, upstream changes and new constraints. Keep increments useful and scoped. Coordinate other owners through linked tasks and exclusive worktrees. Local PR closure does not erase remaining adoption or duplicate retirement.

Tracera: one product, bounded workstreams

Seat	Scope	Outcome
1	Product/schema/integration	Accepted model and thin witness
2	Source inventory/indexing	Actual anchors and supported extraction
3	Intent/spec ingestion	Provenance and supersession
4	Persistence/reconciliation	Durable model and recovery
5	Impact/dissatisfaction	Genuine gaps and uncertainty
6	Assurance ingestion	Native evidence and freshness
7	Product exploration UI	Bounded readable graph/search
8	CLI/MCP/API	Same semantics and permissions
9	Packaging/runtime	Actual install and operational proof
10	Independent evaluator	Baseline and adversarial verification

Integrate before multiplying abstractions

Agree a minimal entity/source/evidence contract first. The existing ALM-oriented atlas crate can be reused where useful, but it must not redefine the product as work bookkeeping. One integration owner and independent verifier accept the combined result.

First witness

Import a real product, trace a capability into source and obligations, expose a real discrepancy, update from accepted evidence and survive restart. A hardcoded graph demo does not establish this outcome.

Capacity is not execution concurrency

Ten plus two for each other 22 products means 54 potential product sessions. Build, GPU, browser/native capture, external quota and review capacity need separate limits. Protect foreground games and live audio.

Prove a reason to keep the custom implementation

Three claim levels

A design can have a plausible advantage; a measured implementation can outperform a defined baseline; a product recommendation additionally needs reliable installation, UX, recovery and maintenance. Do not promote these levels silently.

Fair experiment

Use competent direct, upstream and composed baselines. Separate controlled replay, matched live runs and idiomatic whole-stack comparisons. Record exact models, versions, tasks, hardware, caches, budgets, failures and interventions. Predeclare critical invariants and meaningful margins.

Writing ergonomics

Measure concepts learned, undocumented behavior, glue, errors, debugging, testing and change amplification. Add approval, a new tool, persistence, provider replacement, cancellation and a framework upgrade after the first version works. LOC is supporting context, not the answer.

Agentora

First recover the actual surviving framework API; Cmdra and historical templates using agentkit are not sufficient. Compare direct SDK, OpenAI Agents SDK, LangGraph, CrewAI, Mastra, PydanticAI and a relevant Rust baseline such as Rig. No framework is predeclared the winner.

Accept negative results

A maintained dependency, small wrapper or narrow patch set may be the best outcome. A legitimate internal/fork contract can justify retained code without invented public novelty. The study must retain failed, inconclusive and disadvantageous results.

Research scope

The reference URLs and subject comparators are seeds, not a fresh SOTA review. Expand relevant longlists before architecture decisions and run a justified smaller executable shortlist. Do not pad to a count or claim globally exhaustive search.

Tracera

Persistent product/system model and dissatisfaction discovery

Atlas boundary

Product identity versus repository, work item and session identity; Versioned capability, requirement, interface and evidence relationships; Source anchors, index freshness and build-profile reachability; Bounded expansion/collapse and explainable search.

Current question, not a fresh verdict

Earlier descriptions foreground audit/ALM rather than product semantics; Existing tracera-atlas is ALM-oriented; reuse must not redefine the product. Revalidate the inherited baseline at the current subject revision.

Pilot

Import three representative owned products with seeded and retained-real gaps: an unreachable public capability, stale coverage evidence, a changed dependency, a duplicated owner and a legitimate exception. Ask an unfamiliar operator to explain current behavior, locate the issue, identify affected consumers and plan a bounded repair. Repeat after a source change and restart.

Negative cases

Stale revision is presented as current; Unsupported dynamic edge is asserted as proven; Replayed ingestion duplicates entities; Untrusted repository prose becomes a command.

Measure

Correctness and precision/recall on adjudicated gap cases; Time and interactions to an evidence-backed explanation; Unsupported-claim rate and abstention quality; Incremental update latency and stale-result exposure; Index/model memory and query latency.

Advantage to prove

One persistent explorable product model reduces manual reconciliation across intent, code, contracts and evidence while preserving explainability.

CVP evidence

An installed or hosted real frontend imports an actual product, exposes a genuine discrepancy with source evidence, supports bounded navigation and survives restart without changing ownership or inventing results.

Full editable plan: [products/Tracera/DOSSIER.md](#) | Comparative execution is planned, not completed.

AgilePlus

Repository-scoped governed specification and work tooling

Atlas boundary

Repository-root resolution and isolation; Authoritative state versus generated review files; Transitions, prerequisites, permissions and receipts; CLI/MCP/API/desktop contract equivalence.

Current question, not a fresh verdict

Prior workflow advertised 85% while enforcing different language policies; Tool repository must not own subjects cross-project data. Revalidate the inherited baseline at the current subject revision.

Pilot

Two independent sample products execute a feature from clarified intent through implementation, verification and release preparation. Interrupt the process, change acceptance criteria, replay a request and perform concurrent work. Compare operator steps and recoverability without requiring an ecosystem-wide server.

Negative cases

Wrong repository root selected; Duplicate transition after a restart; Stale acceptance criteria used; Failed native check treated as pass.

Measure

Correct transitions and invalid transitions rejected; Isolation leaks; Recovery and replay behavior; Operator/tool calls per completed feature; Migration burden.

Advantage to prove

Explicit machine-verifiable work contracts remove coordination ambiguity with less administrative effort than ad hoc files and boards.

CVP evidence

Clean installation and an actual scoped CLI/MCP/desktop journey with durable correct state, required checks and a source-bound receipt; unrelated projects work without Phenotype branding or portfolio data.

Full editable plan: `products/AgilePlus/DOSSIER.md` | Comparative execution is planned, not completed.

HeliosLab

Agent workbench or configuration composite; identity adjudication first

Atlas boundary

Actual executable and build roots; Config/flags/secrets packages versus workbench UI; Persisted session/project configuration ownership; Terminal, agent and desktop integration boundaries.

Current question, not a fresh verdict

Baseline found mixed workbench/configuration descriptions; Prior manifests depended on old clap-ext/observability locations. Revalidate the inherited baseline at the current subject revision.

Pilot

First select the accepted public job. For a workbench, launch two isolated projects and different harnesses, navigate current and archived sessions, inspect a tool error, restart, and resume safely. For configuration tooling, create and change scoped flags/secrets and verify applications observe precisely the intended values. Do not combine incomparable outcomes into one score.

Negative cases

Secret exposed in a preview or log; Wrong project receives config; Stale session becomes writable; Missing external source works only from a warm cache.

Measure

Time to first correct workspace/configuration; State fidelity and isolation; Number of manual context switches; Clear failure diagnosis; Clean-install dependency burden.

Advantage to prove

A coherent local workbench or precise configuration tool reduces context switching and setup, not just rebrands a generic workspace.

CVP evidence

The accepted product identity is explicit and the corresponding real UI/CLI workflow installs, persists, survives restart and protects secrets.

Full editable plan: [products/HeliosLab/DOSSIER.md](#) | Comparative execution is planned, not completed.

ResearchLedger

Local research provenance, retrieval and portable vault

Atlas boundary

Raw capture, normalized records and human annotations; Document IDs, duplicates, versions and citation anchors; Vault storage versus search index; Credential and browser boundaries.

Current question, not a fresh verdict

Prior source was paused while names later changed; Installed application and actual vault compatibility not previously qualified. Revalidate the inherited baseline at the current subject revision.

Pilot

Import a mixed corpus of papers, webpages and local notes with duplicates, amended documents, missing metadata and broken URLs. Create a claim with citations, retrieve it offline, change a source, export the vault, then reimport in a clean environment. Compare actual cleanup effort, citation correctness and retention of annotations.

Negative cases

Changed source overwrites cited text; Duplicate import destroys annotations; Credential appears in export; Partial indexing is reported complete.

Measure

Citation precision and source-anchor fidelity; Import completeness with unknowns visible; Deduplication correctness; Cleanup/annotation effort; Offline query latency.

Advantage to prove

Less source cleanup and more trustworthy source-grounded retrieval while keeping the vault portable and privately owned.

CVP evidence

Actual vault survives clean import/export/reopen and supports an evidence-backed research task without private material leaking or a compulsory cloud service.

Full editable plan: [products/ResearchLedger/DOSSIER.md](#) | Comparative execution is planned, not completed.

SessionLedger

Cross-harness capture, preservation, search and replay view

Atlas boundary

Native raw formats and lossless source capture; Normalization of messages/tools/attachments/branches; Time, ordering, deduplication and missing receipts; Redaction, private search and retention.

Current question, not a fresh verdict

Previous pause/active naming contradictions require scoped acceptance; A trace platform is not automatically a lossless arbitrary-transcript archive. Revalidate the inherited baseline at the current subject revision.

Pilot

Archive long interrupted and branched sessions from several harnesses, including duplicate journals, malformed records and missing tool receipts. Search for the actual approval that authorized one action, show the sequence, export it and restore in a clean installation. Introduce a parser upgrade and compare fidelity.

Negative cases

Duplicate journal inflates activity; Missing tool output is fabricated; Attachment orphaned; Replay executes an old tool command.

Measure

Lossless raw preservation and parse coverage; Correct sequence/branch reconstruction; Query correctness; Recovery fidelity; Storage/index overhead.

Advantage to prove

Cross-harness provenance and recovery reduce operator uncertainty without turning transcripts into a new authority or side-effect replay engine.

CVP evidence

An installed consumer ingests real authorized captures, exposes unknowns, retrieves a source-linked event, and restores raw evidence without secret leakage or side effects.

Full editable plan: [products/SessionLedger/DOSSIER.md](#) | Comparative execution is planned, not completed.

KCode

Owned jcode-derived coding harness

Atlas boundary

Provider connection/retry state; Session persistence and compaction; Child-agent ownership and cancellation; Filesystem/tool permission envelope.

Current question, not a fresh verdict

Past consolidation transcript exposed unsafe restart and permission interpretations; A stronger model/harness is not a substitute for scoped action authority. Revalidate the inherited baseline at the current subject revision.

Pilot

Run a multi-step repository task with parent and child agents against an isolated fixture repository. Inject stream disconnects, rate limiting, delayed tools, a failed child and a process restart. Resume while a protected file and forbidden repository remain present. Test the actual installed build and updater.

Negative cases

Duplicate mutation after uncertain completion; Compaction drops a critical restriction; Child outlives cancelled parent; Connection liveness mistaken for progress.

Measure

Useful task completion and forbidden-action rate; Memory at fixed workload/concurrency; Manual recoveries; State loss and duplicate side effects; Tail latency and provider error recovery.

Advantage to prove

Sustained useful operation at lower memory and recovery burden, with stronger rather than weaker mutation safety.

CVP evidence

Owned native binary installs and resumes the controlled fault workload with correct state, no unauthorized mutations, and a measured incremental advantage over upstream.

Full editable plan: [products/KCode/DOSSIER.md](#) | Comparative execution is planned, not completed.

HeliosLite

Forgecode-derived coding-agent CLI

Atlas boundary

Actual binary/brand/release aliases; Current versus legacy/foreign sessions; Stream/retry behavior; Workspace isolation and readonly contracts.

Current question, not a fresh verdict

Earlier promotion already merged; do not repeat it; Latest-release absence in older audit is a revalidation item, not permanent status. Revalidate the inherited baseline at the current subject revision.

Pilot

Use the same long-running coding and transport-fault corpus as KCode. Add legacy/foreign sessions and switch project roots. Search and inspect them without mutating forbidden stores; interrupt a child task and restart the parent. Install and update the fork outside the checkout.

Negative cases

Foreign session becomes writable; Search crosses private workspace boundary; Retry repeats a completed tool; Provider disconnect loses state.

Measure

Completion and recovery rate; Forbidden writes; Session search correctness; Latency tails under disconnect/reconnect; Operator intervention count.

Advantage to prove

Connection continuity and usable session recovery on the actual endpoints, while preserving scope and provenance.

CVP evidence

One current installed fork completes the controlled coding journey, with stable reconnection and safe legacy-session behavior proved at its artifact revision.

Full editable plan: [products/HeliosLite/DOSSIER.md](#) | Comparative execution is planned, not completed.

HeliosCLI

Codex-derived CLI and reusable harness workspace

Atlas boundary

Root workspace versus vendored agent build; Advertised command and actual executable; Auth/provider adapters; Permission and tool boundaries.

Current question, not a fresh verdict

Prior root build excluded vendored Codex; compile success was not agent proof; Missing parent product clarity must be resolved before comparison. Revalidate the inherited baseline at the current subject revision.

Pilot

Build the actual advertised executable in a clean environment, install it and perform a fixed coding task. Trigger a tool failure, denied filesystem action, interruption and session resume. Upgrade the fork and confirm identity and persisted session compatibility.

Negative cases

Wrong workspace builds while binary is missing; Permission context lost on resume; Authentication/config path differs from docs; Model swap changes tool semantics silently.

Measure

Task success and restriction compliance; Setup and maintenance effort; Provider portability; Resumption correctness; Artifact size/startup/resource use.

Advantage to prove

A small useful compatibility or workflow delta that justifies maintaining the fork; inherited Codex features alone are not novelty.

CVP evidence

The installed advertised command, not a sibling test binary, completes an allowed task and handles errors/resume without scope loss.

Full editable plan: [products/HeliosCLI/DOSSIER.md](#) | Comparative execution is planned, not completed.

Khostty

Owned terminal fork or embedded terminal surface

Atlas boundary

Terminal parser/state machine; Renderer/window/input lifecycle; PTY/shell integration and environment; Embedded-library contracts.

Current question, not a fresh verdict

Large upstream fork entails update burden; Novelty must be in owned deltas, not inherited GPU/native behavior. Revalidate the inherited baseline at the current subject revision.

Pilot

Run a terminal conformance/workload corpus: Unicode, wide/combining characters, resize/reflow, alternate screen, bracketed paste, IME, modifiers, high output, screen-reader/keyboard navigation and embedded use where promised. Compare both correctness and perceived/recorded latency at the same workload.

Negative cases

Input lost on focus transition; Escape sequence corrupts state; Resize loses content; Paste executes unexpectedly.

Measure

Conformance failures; Input/render latency and frame pacing; Memory after long sessions; Accessibility and native behavior; Integration effort for owning applications.

Advantage to prove

A specific embedding/workflow improvement with no unacceptable terminal correctness regression.

CVP evidence

An actual installed/embedded terminal performs the claimed delta and passes the relevant correctness corpus against a known upstream reference.

Full editable plan: [products/Khostty/DOSSIER.md](#) | Comparative execution is planned, not completed.

ShareCLI

OS-adjacent process and resource supervisor

Atlas boundary

Process identity and lifecycle; Coalescing equivalence and cache keys; Resource admission/throttling; Cancellation/failure/restart.

Current question, not a fresh verdict

Duplicate source ownership was observed in PhenoTooling; Current source must not inherit an old release validation. Revalidate the inherited baseline at the current subject revision.

Pilot

Run compiler/test/agent workloads with and without ShareCLI, including truly identical pure work and deceptively similar non-coalescible jobs. Introduce input/config/credential differences, cancel a waiter, fail the producer and restart the supervisor. Maintain foreground game/audio workload constraints.

Negative cases

Improper coalescing shares data or skips side effects; Cancelled client terminates other consumers incorrectly; Dead child/resource leak; Cache pollution across subjects.

Measure

Correctness before savings; Duplicate work avoided; Foreground p95/p99 disruption; Memory/CPU/I/O pressure; Cancellation/restart fidelity.

Advantage to prove

Safely remove redundant work and contention without corrupting task semantics or degrading foreground use.

CVP evidence

One installed supervisor handles the complete adversarial workload under realistic contention, with exact behavior and meaningful resource benefit measured.

Full editable plan: [products/ShareCLI/DOSSIER.md](#) | Comparative execution is planned, not completed.

BytePort

Git-to-service deployment application

Atlas boundary

Source ingestion, auth and build inputs; Manifest interpretation and runtime provisioner; Deployment state and ownership; Health/logs/update.

Current question, not a fresh verdict

Earlier docs referenced missing/old NVMS sources; UI startup does not prove deployment. Revalidate the inherited baseline at the current subject revision.

Pilot

Deploy one stateless and one small stateful application through the actual supported target. Change configuration, publish a bad image, fail readiness, introduce a data migration and roll back according to the declared policy. Use a clean machine without private sibling checkouts.

Negative cases

Partial provision reported healthy; Credentials leak to build logs; Retry duplicates resources; Bad image leaves service down without recovery.

Measure

Time/actions to working service; Recovery correctness; Manual interventions; Resource and deployment cost; Clean reproducibility.

Advantage to prove

Less operator work for the local/native or heterogeneous deployment case while maintaining recovery correctness.

CVP evidence

Actual service reachable through the authorized route, configuration and update behavior correct, failed deployment recoverable, data preserved and rollback independently demonstrated.

Full editable plan: [products/BytePort/DOSSIER.md](#) | Comparative execution is planned, not completed.

PhenoFabric

Distributed real-time compute, data and I/O product

Atlas boundary

Logical endpoint/capability versus placement; Same-process/OS/VM/shared-memory/PCIe paths; LAN/WAN transport and memory residency; Seat/focus/surface/audio state.

Current question, not a fresh verdict

Previous work included contract-only transport slices; Codec and hardware assumptions from old docs require verification. Revalidate the inherited baseline at the current subject revision.

Pilot

Desk-to-laptop-to-desk operation with one real application and two actual nodes. Then add a same-host VM route. Keep keyboard/mouse ownership exclusive, switch presentation/audio, transfer one clipboard/file item and recover from a disconnect. Measure each locality cohort separately while background build/inference load runs.

Negative cases

Stale input lease causes duplicate control; Audio feedback or unbounded buffering; Secret clipboard delivered to wrong session; Unknown capability accepted.

Measure

User steps and time to restore workspace; Capture-to-display boundary plus end-to-end separately; Latency distribution and frame pacing; Audio underruns/drift; Color/HDR fidelity and supported fallbacks.

Advantage to prove

A unified user workflow with topology-aware efficient placement, not merely a new transport protocol or a collection of wrapped tools.

CVP evidence

A real two-node public workflow plus a justified local path completes with truthful quality/performance evidence and safe recovery; paper interfaces alone do not qualify.

Full editable plan: [products/PhenoFabric/DOSSIER.md](#) | Comparative execution is planned, not completed.

Pine

Bounded cross-platform application compatibility research

Atlas boundary

Binary/ABI/API support set; Instruction translation versus OS API adaptation; Loader/linker/resource model; Host isolation.

Current question, not a fresh verdict

Broad Wine-equivalent language exceeded prior demonstrated state; No generic compatibility score can replace a program corpus. Revalidate the inherited baseline at the current subject revision.

Pilot

Select a small fixed program corpus with declared binaries, imports and target architecture. Include one useful program beyond trivial startup. Compare observable results, errors, supported calls, overhead and installation with the matching reference path; keep unsupported cases explicit.

Negative cases

Unsupported syscall returns success; ABI mismatch corrupts memory; Isolation escape; Wrong time/filesystem semantics.

Measure

Correct observable results; Explicit unsupported coverage; Startup/runtime overhead; Host resource use; Debuggability.

Advantage to prove

A narrow practical compatibility capability or integration improvement; a general replacement claim requires a much larger corpus.

CVP evidence

One useful declared program runs correctly with bounded failures and a reproducible comparison; research limits remain prominent.

Full editable plan: [products/Pine/DOSSIER.md](#) | Comparative execution is planned, not completed.

OmniRoute

Owned multi-provider model gateway and routing fork

Atlas boundary

Upstream versus owned providers/policies; Credential/session contracts; Retry/fallback/streaming behavior; Quota windows and scopes.

Current question, not a fresh verdict

Upstream submissions are different from owned deployment; Provider breadth does not prove correctness on used endpoints. Revalidate the inherited baseline at the current subject revision.

Pilot

Exercise a fixed used-provider set with normal streams, tool calls, cancellation, partial responses, throttles, auth expiry and hard failures. Compare routing choices under a fixed quality/cost workload. Run a real deployed owned artifact separately from the upstream-contribution queue.

Negative cases

Retry after partial result silently duplicates work; Budget scope wrong; Provider error normalized to false success; Secret logged.

Measure

Task quality held constant; Availability and error fidelity; p95/p99 request overhead; Cost/quota accounting accuracy; Operator recovery burden.

Advantage to prove

A specific operational fit on the actual endpoints with reliable normalization and accounting, not a provider-count contest.

CVP evidence

Deployed fork identity and actual used-provider contracts are qualified with negative cases, accurate accounting and safe cancellation.

Full editable plan: [products/OmniRoute/DOSSIER.md](#) | Comparative execution is planned, not completed.

Substrate

Model routing and/or agent-process execution substrate; boundary split required

Atlas boundary

Model request versus child-process request semantics; Budget and resource ownership; Streams/events/cancellation; Persistence and retry.

Current question, not a fresh verdict

Current naming may conflate two different products; Must not duplicate OmniRoute provider/price authority without a contract. Revalidate the inherited baseline at the current subject revision.

Pilot

Benchmark two distinct workloads rather than a composite score. Model traffic: stream, failover, cancel and account. Process dispatch: spawn, observe, fail, restart and terminate actual tools under explicit scopes. Compare each with the simplest appropriate existing implementation.

Negative cases

Model retry rules incorrectly applied to side-effecting processes; Cancellation loses ownership; Output crosses tenants; Timeout hides a still-running child.

Measure

Correct task semantics; Latency/throughput/resource overhead; Embedding ergonomics; State/recovery correctness; Dependency and patch burden.

Advantage to prove

A precise control/embedding contract with less integration burden; generic routing alone is already well served.

CVP evidence

The chosen public boundary completes real process or model work, with cancellation/auth/budget semantics independently verified and overlapping authorities resolved.

Full editable plan: [products/Substrate/DOSSIER.md](#) | Comparative execution is planned, not completed.

PhenoMLX

Owned MLX/OMLX inference extension and research product

Atlas boundary

Actual inference paths versus experiments; Model/quantization/backend compatibility; Memory/cache lifecycle; Application injection versus self-contained package.

Current question, not a fresh verdict

Prior installation required local oMLX app and adjacent code; A faster result can be a different quantization or cached workload. Revalidate the inherited baseline at the current subject revision.

Pilot

On the same Apple hardware, load the same supported model/quantization and run cold/warm request sets at declared concurrency. Measure quality on fixed prompts, latency distributions and memory. Cancel, unload, restart and create controlled memory pressure; test the actual install recipe on a clean environment.

Negative cases

OOM returns corrupted output; Cancelled request persists indefinitely; Wrong cache reuse; Model unload leaks residency.

Measure

Quality/non-inferiority first; TTFT and inter-token latency distributions; Sustained tokens/s at declared concurrency; Peak/residual memory; Time to clean install.

Advantage to prove

A reproducible quality-preserving throughput, memory or workflow improvement over the actual upstream baseline.

CVP evidence

Clean install, useful inference, cancel/unload/restart and memory pressure behavior are qualified; model and build identity accompany every performance claim.

Full editable plan: [products/PhenoMLX/DOSSIER.md](#) | Comparative execution is planned, not completed.

Portage

Harbor-derived evaluation runner and environment adapters

Atlas boundary

Task/environment contracts; Fork delta and install source; Agent adapters and isolation; Reward/oracle execution.

Current question, not a fresh verdict

Old instructions used TEMP repository names; Metadata or suite-info commands are not execution evidence. Revalidate the inherited baseline at the current subject revision.

Pilot

Run one known-good and one known-bad agent through a real environment/task. Add an infrastructure failure, timeout, missing dependency and cancellation. Compare the fork with its upstream on retained output fidelity, configuration effort and any owned adapter/scheduling improvement.

Negative cases

Missing implementation skips to green; Reward fabricated from metadata; Sandbox not used by tool code; Logs lost after cancellation.

Measure

Correct outcome classification; Reproducibility; Setup and adapter authoring effort; Resource enforcement; Artifact completeness.

Advantage to prove

A measured environment/adapter/scheduling or evidence benefit over upstream, not a renamed runner.

CVP evidence

Clean fork installation runs the useful benchmark and deliberately failing cases, preserving logs/rewards and enforcing declared limits.

Full editable plan: [products/Portage/DOSSIER.md](#) | Comparative execution is planned, not completed.

PhenoLab

Reproducible experiment/evaluation consumer

Atlas boundary

Datasets and source provenance; Policies/solvers/scorers; Model and environment configuration; Oracles and failure classification.

Current question, not a fresh verdict

Prior negative oracle skipped based on implementation patterns; Runner presence does not establish benchmark execution. Revalidate the inherited baseline at the current subject revision.

Pilot

Reproduce one useful experiment through both PhenoLab and a direct baseline. Freeze inputs, models, resource limits and scoring. Include bad answers, malformed output, missing environment, known task failure and infrastructure failure. Repeat on a clean host or independent operator where feasible.

Negative cases

Unimplemented feature determines whether its own test runs; Unsourced case counted as success; Model failure mixed with infrastructure outage; Only selected good trials retained.

Measure

Reproduction success; Scorer validity and calibration; Failure classification accuracy; Operator effort; Total evaluation cost.

Advantage to prove

Less work to produce trustworthy reusable experiments, with stronger failure accounting rather than more abstractions.

CVP evidence

One real reproducible experiment bundle includes all outcomes, source/config identities, actual scorer evidence and an independent known-negative control.

Full editable plan: [products/PhenoLab/DOSSIER.md](#) | Comparative execution is planned, not completed.

Civis

Persistent game-first systemic sandbox

Atlas boundary

Player roles, controls and authority transitions; Renderer/assets/animation/audio versus headless core; World generation, multipart seeds and checkpoint branching; Knowledge/technology/life systems and causal dependencies.

Current question, not a fresh verdict

Old global determinism/research-only language conflicts with current intent; Minimal presentation build is not the game. Revalidate the inherited baseline at the current subject revision.

Pilot

Use one living settlement with a visible water/food constraint. Intervene as god, redirect a leader-level action, inhabit an affected person, save/quit/reopen and continue. Leave a bounded background run and inspect source-backed changes. Separately compare rendering/authoring paths on the same checkpoint and content target.

Negative cases

Role switch loses identity or intent; Missing required asset silently becomes primitive; Save omits cultural/social state; Background time jump destroys world.

Measure

Playable coherent journey and qualitative clarity; Frame pacing and interaction latency; Visual/audio acceptance under supported profiles; Save/recovery fidelity; Overnight resource/time behavior.

Advantage to prove

A convincing persistent world with fluid but meaningful god/leader/individual play and understandable systemic consequences.

CVP evidence

Installed player build outside the checkout visibly supports the accepted journey, required presentation, actual state continuity and bounded unattended execution; headless tests alone are insufficient.

Full editable plan: [products/Civis/DOSSIER.md](#) | Comparative execution is planned, not completed.

Dino

Game mod/pack platform and companion tooling

Atlas boundary

Mod bootstrap and upstream injector; Pack manifests and dependency resolution; SDK and game bridge APIs; Asset build/import and runtime loading.

Current question, not a fresh verdict

A schema/SDK build does not prove the mod loads in the supported game; Fork licensing and update compatibility remain part of the contract. Revalidate the inherited baseline at the current subject revision.

Pilot

Author the same small visible gameplay/content pack manually and using DINOForge. Add an asset, change behavior, create a conflicting pack and unsupported version, reload where promised, restart the game, update and uninstall. Capture the actual game and companion UI.

Negative cases

Pack accepted but not loaded; Missing asset silently ignored; Mod conflict corrupts save; Game update breaks injector.

Measure

Time to first working pack; Iteration and diagnosis effort; Compatibility/failure isolation; Runtime overhead; Asset quality and fidelity.

Advantage to prove

A safer and faster practical mod-authoring workflow with real game integration and useful diagnostics.

CVP evidence

A new pack visibly works in the installed supported game, handles conflicts and failures correctly, and can be disabled/uninstalled without corrupting owned or unowned state.

Full editable plan: [products/Dino/DOSSIER.md](#) | Comparative execution is planned, not completed.

WorldSphereMod

Game-specific spherical/3D presentation mod

Atlas boundary

Actually tested branch and fork delta; Render hooks and required asset bundles; LOD/camera/world representation; Game compatibility and other mods.

Current question, not a fresh verdict

Broad game infrastructure may live outside this repo; A quiet default branch does not prove worker inactivity. Revalidate the inherited baseline at the current subject revision.

Pilot

Load the same populated world under upstream and the candidate. Traverse camera and LOD ranges, inspect animation and terrain, save/load and disable the mod. Repeat with the declared compatible mod set and on the supported graphics profiles.

Negative cases

LOD pops or resource duplication; Save incompatibility; Camera/input mismatch; Required bundle missing.

Measure

Visual correctness and readability; Frame-time distribution; Memory and loading stalls; Compatibility pass/fail; Recovery/disable behavior.

Advantage to prove

The specific 3D/world presentation improvement with explicit acceptable performance and compatibility trade-offs.

CVP evidence

Qualified package renders the real game under declared profiles, preserves save behavior and has a demonstrated safe removal path.

Full editable plan: [products/WorldSphereMod/DOSSIER.md](#) | Comparative execution is planned, not completed.

CivicWarfare

Distributed game/mod scenario with potentially private producer tools

Atlas boundary

Public client versus private production sources; Distributed artifact and game/version compatibility; Gameplay/control/information loop; Telemetry consent and persistence.

Current question, not a fresh verdict

Public snapshot may intentionally omit generators; Repository rename does not change product obligations. Revalidate the inherited baseline at the current subject revision.

Pilot

Install the distributed mod/scenario as a new user, understand objectives and controls, complete a representative gameplay loop, save/load, experience a failure and remove/update the package. Compare onboarding, readability and behavior with the closest real baseline.

Negative cases

Undeclared private dependency required at runtime; Telemetry without consent; Misleading objective/UI state; Save/version mismatch.

Measure

First useful/understood play time; Gameplay and qualitative clarity; Performance under intended scale; Install/update/save fidelity; Support/recovery burden.

Advantage to prove

A distinct enjoyable scenario or gameplay experience; novelty need not be in databases/renderers underneath it.

CVP evidence

The distributed public artifact installs and plays as intended, with authentic evidence, correct save/version handling and clear producer provenance.

Full editable plan: [products/CivicWarfare/DOSSIER.md](#) | Comparative execution is planned, not completed.

Melosviz

Music visualization authoring and rendering application

Atlas boundary

Audio import/analysis and timing; Scene/preset generation versus demo fixtures; Real render backend and editor; Native/web shell and optional external services.

Current question, not a fresh verdict

Prior production references included backend stubs; Generating a ZIP does not prove media quality. Revalidate the inherited baseline at the current subject revision.

Pilot

Use the same authorized music excerpt to import, analyze, play/synchronize, edit a scene, render/export and reopen. Then replace the track, change timing, use a new preset, disable a backend and recover. Compare actual output and editing effort, not only generation latency.

Negative cases

Stub backend produces plausible but wrong output; A/V drift; Render cancelled but resources remain; Export missing codec/frames.

Measure

Time to useful visualization and revision; A/V synchronization error; Frame pacing and render duration; Output correctness and visual quality; Recovery and project portability.

Advantage to prove

A faster music-specific path to controllable, repeatable visual work with reliable output, not generic graphics novelty.

CVP evidence

An actual installed or deployed application handles real media through the complete authoring/export/reopen path with measured synchronization and truthful backend status.

Full editable plan: [products/Melosviz/DOSSIER.md](#) | Comparative execution is planned, not completed.

Pooled foundations

PhenoInfra / pooled-foundation

Recover runtime/provisioner, manifest, secrets, state, backup/restore and consumer contracts. Separate reusable libraries from environment-specific resources. A real clean consumer provisions and recovers its intended resource, and a restore drill preserves state and ownership.

PhenoTooling / pooled-foundation

Inventory all languages and nested workspaces, advertised commands, native tools, copied products and false-green scaffolds. Every verifier must prove a negative case. Real checkers reject seeded failures; product copies have one writer/release authority; actual consumers install and use the supported package.

PhenoAI / pooled-foundation

Identify embeddings, providers, agent SDK, device drivers and artifact contracts independently. Resolve which consumers require each package and distinguish Agentora from similarly named CLI frameworks. Each retained package has a clean consumer, truthful support profile and an adopt/patch/wrap/retain decision backed by tests.

PhenoGfx / pooled-foundation

Map data ownership, renderer adapters, native ABI, asset conversion and required game consumers. A byte-compatible API and a semantically correct rendering path are different proofs. At least the accepted real consumers exercise the canonical package with correct ABI/data/rendering and explicit performance budgets.

These are capability/consumer obligations, not automatically separate product seats. Source/target migrations require accepted scope and authority.

Supporting and conditional targets

PhenoRegistry / supporting-surface

Use immutable repo IDs and explicit accepted lifecycle/role records. Separate GitHub facts from proposed disposition and externally owned work/evidence state. The live roster resolves aliases, names current allowed actions and links truthful evidence without overriding source owners.

PhenoDesign / supporting-surface

Locate editable source, export recipes, actual consumers and accepted brand scope. Do not regenerate a brand because a legacy README says archived. Real consumers use consistent tokens/assets; visual, accessibility, performance and rights checks pass; source edits reproducibly produce deliverable variants.

KooshaPari / supporting-surface

Map each public claim to an actual product/release, active URL, docs version and artifact. Preserve private/public boundaries and honest installation status. An unfamiliar visitor can understand the portfolio, find working product/docs/install paths and distinguish measured capability from plans.

PhenoApps / conditional-target

Resolve the actual FocalPoint application source, Localbase and RIPFitnessApp future disposition separately. A shared repository must not force a shared product, database or release lifecycle. Each accepted app remains independently installable, testable and understandable; no archive or donor deletion follows from this dossier.

Agentora-capability / absorbed-capability

Inventory the actual exported Rust/other framework, tools, state, model adapters, storage and consumers. Locate legacy Agentora code by lineage rather than an agentkit string alone. A matched coding-agent pilot proves correctness, ergonomics and recovery; retained custom behavior has a defensible measured benefit or internal boundary.

These are capability/consumer obligations, not automatically separate product seats. Source/target migrations require accepted scope and authority.

Use the package without overstating it

What has been created

Editable project-specific dossiers, pilot plans, proposed program contracts, provisional feature bundles, schemas, task graph, agent assignments, synthetic examples, reference checkers and tests. Original contextual archives and exact visible human requests are preserved with hashes.

What has not been performed

No fresh complete GitHub audit, native product suite, installed product inspection, SOTA benchmark, migration, release, DNS/provider change or actual work-engine transition. No product superiority or completion is certified.

SROC/CDP

The exact terms are unresolved. Recover their original definitions and map them deliberately. Do not invent an acronym expansion or let that block unambiguous atlas and assurance progress.

Validation scope

See VALIDATION_REPORT.md and validation/ for actual executed results. Schema and linkage checks do not prove semantic completeness. The evidence checker is a reference integrity layer, not an authenticated approval engine.

Next action

Give each current pair its relevant dossier and prompt. Bind source, scope, actual artifact, critical obligations and first useful target. Keep normal work moving; choose mature reuse when evidence favors it; close a real current viable product instead of generating another unbounded planning corpus.